Example

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Problem Statement

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Problem Statement: 4-Element Uniform Linear Array

Consider a uniform linear array (ULA) consisting of $N = 4$ isotropic sensors located along the z-axis. The interelement spacing is $d = \lambda/2$, where λ is the wavelength of the incoming narrowband plane waves. The array is centered at the origin so that the element positions are symmetric about $z = 0$. For simplicity, set $\lambda = 1$ (arbitrary units).

Two plane waves of the same frequency arrive from different directions:

- **Direction A (broadside):** $\theta = 90^\circ$ (i.e., perpendicular to the array axis)
- **Direction B (off-broadside):** $\theta = 45^\circ$ (measured from the positive z-axis)

The wavenumber component along the z-axis is given by $k_z = -\frac{2\pi}{\lambda}\cos\theta$.

Answer the following questions:

1. **Array Positions**
   Determine the coordinates p_{z_n} of the four elements using the centering formula

$$p_{z_n} = \left(n - \frac{N-1}{2}\right) d, \quad n = 0, 1, 2, 3.$$

2. **Manifold Vectors**
   For each direction, compute the array manifold vector $\mathbf{v}(k_z) = \left[e^{-jk_z p_{z_n}}\right]_{n=0}^{3}$.
   - (a) For Direction A ($k_z = 0$).
   - (b) For Direction B (exact numerical values, keeping at least four decimal places). Verify the conjugate symmetry property for Direction B.
3. **Conventional Beamformer Output (Uniform Weights)**
   A conventional beamformer is implemented with uniform weights $w_n = 1/N$ and steered to broadside.
   - (a) Write the weight vector $\mathbf{w}$.
   - (b) Compute the array output $y = \mathbf{w}^H \mathbf{v}$ for each of the two incident plane waves.
   - (c) Confirm that the output for the broadside signal equals 1, and interpret the result for the off-broadside signal as the value of the beam pattern at $\theta = 45^\circ$.

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4. **Steering (Delay Compensation)**
   - (a) Determine the weight vector $\mathbf{w}_{45}$ required to steer the main response axis to Direction B ($\theta = 45^\circ$).
   - (b) Verify that this weight vector satisfies the distortionless constraint $\mathbf{w}_{45}^H \mathbf{v}(\text{Dir B}) = 1$.
   - (c) Compute the response of this steered beamformer to a signal arriving from broadside (Direction A) and comment on the result.
5. **Performance Measures for the Broadside-Steered Array**

   For the uniform weight vector $\mathbf{w}$ from question 3, compute:
   - (a) The white noise array gain $A_w = 1/\|\mathbf{w}\|^2$.
   - (b) The directivity D (recall that for a standard linear array with $d = \lambda/2$, $D = N$).
   - (c) The sensitivity function $T_{\text{se}} = \|\mathbf{w}\|^2$.
   - (d) The approximate half-power beamwidth (HPBW) in u-space (Δu) and in θ-space at broadside. Use the approximation $\Delta u \approx 0.891 \frac{\lambda}{Nd}$.

6. **Grating Lobes**

   Explain why, with $d = \lambda/2$, no grating lobes appear in the visible region for any steering angle within $-90^\circ \leq \theta \leq 90^\circ$.

All numerical answers should be given with sufficient precision (e.g., four decimal places for complex quantities). This problem illustrates the fundamental concepts of array geometry, manifold vectors, beamforming, steering, and performance measures as introduced in Chapter 2.

1. Array Position

$$P_{z_n} = \left(n - \frac{N-1}{2}\right) d \qquad n = 0, 1, \ldots, N-1$$

for $N = 4$; $d = \frac{\lambda}{2} = \frac{1}{2} = 0.5$

$$P_{z_n} = (n - 1.5) \times 0.5 \qquad n = 0, 1, 2, 3$$

$n = 0 \rightarrow P_{z_0} = (0 - 1.5)(0.5) = -0.75$

$n = 1 \rightarrow P_{z_1} = (1 - 1.5)(0.5) = -0.25$

$n = 2 \rightarrow P_{z_2} = (2 - 1.5)(0.5) = 0.25$

$n = 3 \rightarrow P_{z_3} = (3 - 1.5)(0.5) = 0.75$

thus :

$$\vec{P} = [-0.75, -0.25, 0.25, 0.75]^T$$

(in unit of λ).

Manifold vector A

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a) Direction of A (Broadside, $\theta = 90^\circ$)

wavenumber is given by

$$k_z = -\frac{2\pi}{\lambda}\cos\theta$$

$\theta = 90 \rightarrow \cos 90 = 0 \rightarrow k_z = 0$

The array manifold vector is

$$\vec{V_A} = \begin{bmatrix} e^{-jk_z P_{zn}} \end{bmatrix} \quad n = 0, 1, 2, 3$$

$k_z = 0$ Then

$$\vec{V_A} = [1, 1, 1, 1]^T$$

b) Direction B ($\theta = 45^\circ$):

$$\cos 45^\circ = \frac{\sqrt{2}}{2} = 0.70710678$$

$$k_z = -\frac{2\pi}{\lambda}\cos\theta \rightarrow \approx -2\pi \times 0.70710678$$

$$k_z \simeq -4.442882938$$

Manifold vector B

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$$\vec{V_B} = \begin{bmatrix} -jk_z P_{zn} \\ e \end{bmatrix} \quad n = 0, 1, 2, 3$$

After computation $\vec{V_B}$ will be obtained

$$\vec{V_B} = \begin{bmatrix} -0.9808 + j\,0.1951 \\ 0.4440 - j\,0.8969 \\ 0.4440 + j\,0.8960 \\ -0.9809 - j\,0.1951 \end{bmatrix}$$

Conjugate symmetry is evident:

$$V_0 = V_3^* \quad \& \quad V_1 = V_2^*$$

3. Conventional Beamformer output (uniform weights, steered to Broadside).

a) weight vector

$$\vec{W} = \frac{1}{N}\vec{1} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}^T$$

b) output for the two plan waves

For direction A:

$$y_A = w^H \vec{v}_A = \frac{1}{4}(1+1+1+1) = 1$$

For direction B:

$$y_B = w^H \vec{v}_B = \frac{1}{4}\Big[(-0.9808 + j0.1951)$$

$$+(0.4440 - j0.8960) + (0.4440 + j0.8960)$$

$$+(-0.9808 - j0.1951)\Big]$$

$$y_B = -1.0736/4 = \boxed{-0.2684}$$

(c) Interpretation

The output for the broadside signal is exactly 1 (distortionless).
The output for the off-broadside signal, $|y_B| = 0.2684$, is the value of the beam pattern $B(\theta)$ at $\theta = 45^\circ$ when the array is steered to broadside.

Steering Direction

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4. Steering to Direction B ($\theta = 45^\circ$)

(a) Steering weight vector

$$\mathbf{w}_{45} = \frac{1}{N}\mathbf{v}_B = \frac{1}{4}\begin{bmatrix} -0.9808 + j0.1951 \\ 0.4440 - j0.8960 \\ 0.4440 + j0.8960 \\ -0.9808 - j0.1951 \end{bmatrix}.$$

(b) Distortionless constraint

$$\mathbf{w}_{45}^H \mathbf{v}_B = \frac{1}{4}\mathbf{v}_B^H \mathbf{v}_B = \frac{1}{4} \times 4 = 1,$$

since $\mathbf{v}_B^H \mathbf{v}_B = \sum_{n=0}^{3} |v_n|^2 = 4$ (each $|v_n|^2 = 1$). The constraint holds.

(c) Response to a broadside signal

$$\mathbf{w}_{45}^H \mathbf{v}_A = \frac{1}{4}\mathbf{v}_B^H \mathbf{1} = \frac{1}{4}(-1.0736) = -0.2684,$$

which matches the magnitude of the earlier broadside-steered response to Direction B. This illustrates the reciprocal nature of the beam pattern: the response when steering to B and looking at A equals the response when steering to A and looking at B.

Performance Measurment

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5. Performance Measures for the Broadside-Steered Array

(a) White noise array gain

$$\|\mathbf{w}\|^2 = \sum_{n=0}^{3} |w_n|^2 = 4 \times \left(\frac{1}{4}\right)^2 = \frac{1}{4}.$$

Thus

$$A_w = \frac{1}{\|\mathbf{w}\|^2} = 4.$$

(b) Directivity

For a standard linear array with $d = \lambda/2$, Eq. (2.160) gives $D = N = 4$.
So $D = 4$, which equals A_w in this case.

(c) Sensitivity function

$$T_{\text{se}} = \|\mathbf{w}\|^2 = \frac{1}{4} = 0.25.$$

(d) Half-power beamwidth (HPBW)

Using the approximation $\Delta u \approx 0.891 \frac{\lambda}{Nd}$ with $\lambda = 1$, $N = 4$, $d = 0.5$:

$$\Delta u \approx 0.891 \frac{1}{4 \times 0.5} = 0.891 \frac{1}{2} = 0.4455.$$

In θ-space at broadside, since $u = \cos\theta$ and near broadside $\Delta\theta \approx \Delta u$ (in radians),

$$\text{HPBW} \approx 0.4455 \text{ rad} \approx 25.5^\circ.$$

Grating Lobes

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6. Grating Lobes

Grating lobes occur when the numerator and denominator of the beam pattern expression (Eq. (2.97)) simultaneously equal one, i.e., when $\frac{\psi}{2} = m\pi$ (m integer) or $\psi = 2m\pi$.
In u-space, $\psi = \frac{2\pi d}{\lambda} u = \pi u$ (since $d/\lambda = 0.5$). The visible region is $|u| \leq 1$.
The condition for a grating lobe is $\pi u = 2m\pi \Rightarrow u = 2m$.
For $|u| \leq 1$, the only possible integer m is 0, which corresponds to the main lobe. Hence no grating lobes appear in the visible region. This holds for any steering direction because steering merely shifts the pattern in u-space, but the spacing $d = \lambda/2$ ensures that the first grating lobe ($u = \pm 2$) remains outside the visible region for all steered angles.